

AMENDMENT NO. 1 NOVEMBER 2012
TO
IS 1786 : 2008 HIGH STRENGTH DEFORMED BARS AND WIRES
FOR CONCRETE REINFORCEMENT — SPECIFICATION

(Fourth Revision)

(Second cover page, Foreword) — Insert the following after sixth para as a separate para:

‘Provisions in this standard are at variance with similar provisions in ISO 6935-2 : 2007 ‘Steel for the reinforcement of concrete — Part 2: Ribbed bars’, in view of the following:

- a) Geographical factors which determine the earthquake zoning and consequently the structural design considerations, structural design method/principles adopted, the design parameters and the required material properties;
- b) Technological factors associated with the process of manufacture of the product (such as through secondary processes/induction furnace), which influence the product characteristics (like chemical composition and mechanical properties like yield strength, tensile strength, elongation, ductility, etc); and
- c) Construction techniques and practices adopted in this country, the equipments used and the skill level of construction workers which also influence the product characteristics (such as bend and bond strength).

The major deviations are:

- a) The steel grades covered are at variance. This standard covers requirements for high strength bars only whereas ISO 6935-2 also covers bars of lower tensile strengths.
- b) Chemical compositions vary from that in ISO 6935-2.
- c) Mechanical properties specified in this standard are individual values, whereas the tensile properties in ISO 6935-2 are primarily the specified characteristic values.
- d) Bend and re-bend test requirements vary from those in ISO 6935-2.
- e) Bond requirements in this standard are specified on basis of mean area of ribs whereas the requirements in ISO 6935-2 are for rib geometry. This standard additionally specifies pull out test as a requirement for approval of new or amended rib geometry.’

(Page 1, clause 1.1) — Substitute the following for the existing clause:

‘**1.1** This standard covers the requirements of deformed steel bars and wires for use as reinforcement in concrete, in the following strength grades:

- a) Fe 415, Fe 415D, Fe 415S;
- b) Fe 500, Fe 500D, Fe 500S;
- c) Fe 550, Fe 550D; and
- d) Fe 600.

NOTES

1 The figures following the symbol Fe indicate the specified minimum 0.2 percent proof stress or yield stress, in N/mm².

2 The letters D and S following the strength grade indicates the categories with same specified minimum 0.2 percent proof stress/yield stress but with enhanced and additional requirements.’

(Page 2, clause 4.2) — Substitute the following for the existing clause:

‘**4.2** The ladle analysis of steel for various grades, when carried out by the method specified in the relevant parts of IS 228 or any other established instrumental/chemical method, shall have maximum permissible percentage of constituents as follows. In case of dispute, the procedure given in IS 228 and its relevant parts

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shall be the referee method and where test methods are not specified shall be as agreed to between the purchaser and the manufacturer/supplier.

<i>Constituent</i>	<i>Percent, Maximum</i>								
	Fe 415	Fe 415D	Fe 415S	Fe 500	Fe 500D	Fe 500S	Fe 550	Fe 550D	Fe 600
Carbon	0.30	0.25	0.25	0.30	0.25	0.25	0.30	0.25	0.30
Sulphur	0.060	0.045	0.045	0.055	0.040	0.040	0.055	0.040	0.040
Phosphorus	0.060	0.045	0.045	0.055	0.040	0.040	0.050	0.040	0.040
Sulphur and phosphorus	0.110	0.085	0.085	0.105	0.075	0.075	0.100	0.075	0.075

NOTES

1 For guaranteed weldability, the Carbon Equivalent, CE using the formula:

$$CE = C + \frac{Mn}{6} + \frac{(Cr + Mo + V)}{5} + \frac{(Ni + Cu)}{15}$$

shall not be more than 0.53 percent, when micro-alloy/low alloys are used. When micro-alloys/low alloys are not used, carbon equivalent using the formula:

$$CE = C + \frac{Mn}{6}$$

shall not be more than 0.42 percent. Reinforcement bars/wires with carbon equivalent above 0.42 percent should, however be welded with precaution. Use of low hydrogen basic coated electrodes with matching strength bars/wires is recommended.

2 Addition of micro-alloying elements is not mandatory for any of the above grades. When strengthening elements like Nb, V, B and Ti are used individually or in combination, the total contents shall not exceed 0.30 percent; in such case manufacturer shall supply the purchaser or his authorized representative a certificate stating that the total contents of the strengthening elements in the steel do not exceed the specified limit.

3 Low alloy steel may also be produced by adding alloying elements like Cr, Cu, Ni, Mo and P, either individually or in combination, to improve allied product properties. However, the total content of these elements shall not be less than 0.40 percent. In such case, the manufacturers shall supply the purchaser or his authorized representative a test certificate stating the individual contents of all the alloying elements. In such low alloy steels when phosphorus is used, it shall not exceed 0.12 percent and when used beyond the limit prescribed in 4.2, the carbon shall be restricted to a maximum of 0.15 percent, and in such case the restriction to maximum content of sulphur and phosphorus as given in 4.2 and the condition of minimum alloy content 0.40 percent shall not apply.

User may note that there is a danger of pitting and crevice corrosion when weathering steels (that is those with chemical composition conforming to IS 11587) are embedded in chloride contaminated concrete.”

4 Nitrogen content of the steel should not exceed 0.012 percent (120 ppm), which shall be ensured by the manufacturer by occasional check analysis. Higher nitrogen contents up to 0.025 percent (250 ppm) may be permissible provided sufficient quantities of nitrogen binding elements, like Nb, V, Ti, Al, etc, are present. In order to ascertain whether sufficient quantities of nitrogen binding elements are present, following formula may be used, where all elements are in ppm.

$$(N - 120) < \frac{Al_{free}}{10} + \frac{(Ti + V)}{7} + \frac{Nb}{14}$$

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(Page 4, clause 6.1) — Substitute the following for the existing clause:

‘6.1 The nominal sizes of bars/wires shall be as follows:

Nominal size, 4 mm, 5 mm, 6 mm, 8 mm, 10 mm, 12 mm, 16 mm, 20 mm, 25 mm, 28 mm, 32 mm, 36 mm, 40 mm, 45 mm, 50 mm.

NOTE — Other sizes may be supplied by mutual agreement.’

(Page 4, Table 1) — Insert the following at the end of the Table 1:

SI No.	Nominal Size mm	Cross Sectional Area mm ²	Mass per Metre kg
(1)	(2)	(3)	(4)
xiv)	45	1591.1	12.49
xv)	50	1964.4	15.42

(Page 6, Table 3) — Substitute the following for the existing Table 3:

**Table 3 Mechanical Properties of High Strength
Deformed Bars and Wires**
(Clause 8.1)

SI No.	Property	Fe 415	Fe 415D	Fe 415S	Fe 500	Fe 500D	Fe 500S	Fe 550	Fe 550D	Fe 600
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	0.2 percent proof stress/ yield stress, <i>Min</i> , N/mm ²	415.0	415.0	415.0	500.0	500.0	500.0	550.0	550.0	600.0
ii)	0.2 percent proof stress/ yield stress, <i>Max</i> , N/mm ²	—	—	540.0	—	—	625.0	—	—	—
iii)	TS/YS ratio ¹⁾ , N/mm ²	≥ 1.10, but TS not less than 485.0 N/mm ²	≥ 1.12, but TS not less than 500.0 N/mm ²	1.25	≥ 1.08, but TS not less than 545.0 N/mm ²	≥ 1.10, but TS not less than 565.0 N/mm ²	1.25	≥ 1.06, but TS not less than 585 N/mm ²	≥ 1.08, but TS not less than 600.0 N/mm ²	≥ 1.06, but TS not less than 660 N/mm ²
iv)	Elongation, percent, min. on gauge length $5.65\sqrt{A}$, where <i>A</i> is the cross-sectional area of the test piece	14.5	18.0	20.0	12.0	16.0	18.0	10.0	14.5	10.0
v)	Total elongation at maximum force, percent, <i>Min</i> , on gauge length $5.65\sqrt{A}$, where <i>A</i> is the cross-sectional area of the test piece (see 3.9) ²⁾	—	5	10	—	5	8	—	5	—

¹⁾ TS/YS ratio refers to ratio of tensile strength to the 0.2 percent proof stress or yield stress of the test piece

²⁾ Test, wherever specified by the purchaser.

(Page 6, clause 9.1.2.1) — Insert the following note at the end of clause:

‘NOTE— In case of any dispute, the results obtained from full bar test pieces (without machining to remove deformations) shall be treated as final and binding.’

(Page 7, clause 9.2.1, line 7) — Substitute ‘Fe 415, Fe 415D and Fe 415S bars/wires’ for ‘Fe 415 and Fe 415D bars/wires’.

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(Page 7, clause 9.2.1, line 8) — Substitute ‘Fe 500, Fe 500D and Fe 500S bars/wires’ for ‘Fe 500 and Fe 500D bars/wires’.

(Page 7, clause 9.3, line 2) — Insert ‘maximum’ before ‘mandrel’.

(Page 7, clause 9.3, Table 4) — Substitute the following for the existing Table 4:

Table 4 Maximum Mandrel Diameter for Bend Test
(Clause 9.3)

Sl No.	Nominal Size mm	Maximum Mandrel Diameter for Different Grades								
		Fe 415	Fe 415D	Fe 415S	Fe 500	Fe 500D	Fe 500S	Fe 550	Fe 550D	Fe 600
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	Up to and including 20	3 ϕ	2 ϕ	2 ϕ	4 ϕ	3 ϕ	3 ϕ	5 ϕ	4 ϕ	5 ϕ
ii)	Over 20	4 ϕ	3 ϕ	3 ϕ	5 ϕ	4 ϕ	4 ϕ	6 ϕ	5 ϕ	6 ϕ

NOTE — ϕ is the nominal size of the test piece, in mm.

(Page 7, clause 9.4.1) — Substitute the following for the existing clause:

9.4.1 The maximum diameter of the mandrel shall be as given below:

Sl No.	Nominal Size of Specimen	Maximum Dia of Mandrel for Fe 415 and Fe 500	Maximum Dia of Mandrel for Fe 415D, Fe 415S, Fe 500D and Fe 500S	Maximum Dia of Mandrel for Fe 550 and Fe 600	Maximum Dia of Mandrel for Fe 550D
(1)	(2)	(3)	(4)	(5)	(6)
i)	Up to and including 10 mm	5 ϕ	4 ϕ	7 ϕ	6 ϕ
ii)	Over 10 mm	7 ϕ	6 ϕ	8 ϕ	7 ϕ

NOTE — ϕ is the nominal size of the test piece, in mm.